

ISOLATION OF OPERATING FIELD IN OPERATIVE DENTISTRY (PART -1)

Learning outcomes

- Analyze the advantages and disadvantages of isolation of operating field
- Summarize the parts of rubber dam and its application



Moisture Control



- Operative dentistry cannot be executed properly unless the moisture in the mouth is controlled.
- Moisture control refers to excluding sulcular fluid, saliva, and gingival bleeding from the operating field.
- The rubber dam, suction devices, and absorbents are variously effective in moisture control.
- Generally, the rubber dam is the recommended technique for moisture control.

RUBBER DAM

- 1864 S C Barnum first described rubber dam
- The use of rubber dam is mandatory in restorative and endodontic procedures.
- The rubber dam helps in eliminating saliva from the operating site and also retracts the soft tissue.



Advantages

Sturdevant 5 th edition Page 463

1. **Dry Clean Operating Field**
2. **Improved Visibility**
3. **Improved Properties of Dental Materials**
4. **Protection of the Patient and Operator**
5. **Operating Efficiency**



Disadvantages

1. Time consuming and patients objection.
2. Unusual tooth shapes or positions that cause inadequate clamp placement

Example-

- ✓ Partially erupted teeth
- ✓ Third molars
- ✓ Malpositioned teeth



3. Patients suffering from Bronchial asthma may not tolerate the rubber dam if breathing through the nose is difficult.
4. Some patients may be allergic to latex making the use of rubber dam impossible. (Latex free rubber can be used)



Parts

- Rubber dam sheets
- Rubber dam Holder
- Rubber ram Clamp or retainer
- Rubber dam Punch
- Retainer Forceps
- Napkin
- Accessories



Rubber Dam Sheets

- Rubber dam is made from natural latex rubber .
- Sizes
 - 5x5 inch
 - 6x6 inch
- Latex free dams are also available.
(In case of latex rubber allergy)

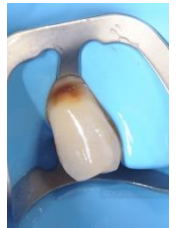




- Rubber dam material, as with all rubber products, deteriorates over time, resulting in low tear strength.
- Material that is reasonably new should be used.
- Light and dark dam materials are available, but the dark color is preferred for contrast.
- Rubber dam material has a shiny and a dull side.
- Because the dull side is less light reflective, it is generally placed facing the occlusal side of the isolated teeth.

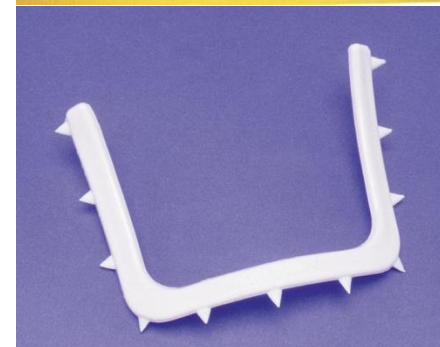
Thickness

- The thickness or weights available are
 - Thin (0.006 inch)
 - Medium (0.008 inch)
 - Heavy (0.010 inch)
 - Extra heavy (0.012 inch)
 - Special heavy (0.014 inch)
- Thicker dam - More effective in retracting gingival tissue, more resisting to tearing ,it is especially recommended for isolating Class V caries lesions in conjunction with a cervical retainer.
- Thinner material – In case of tight contacts



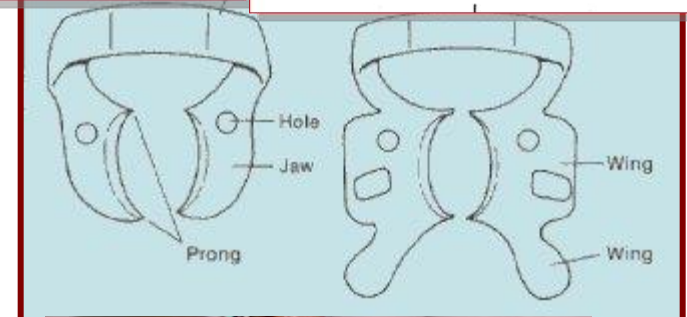
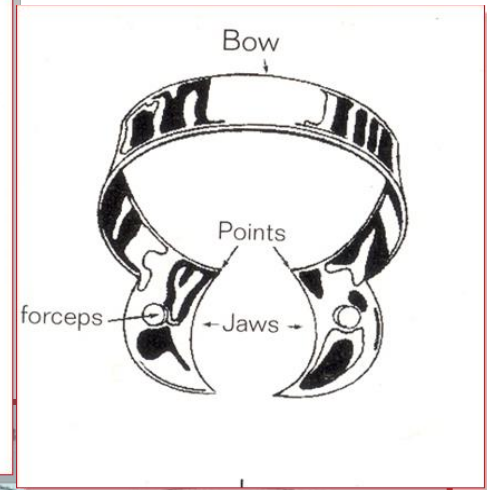
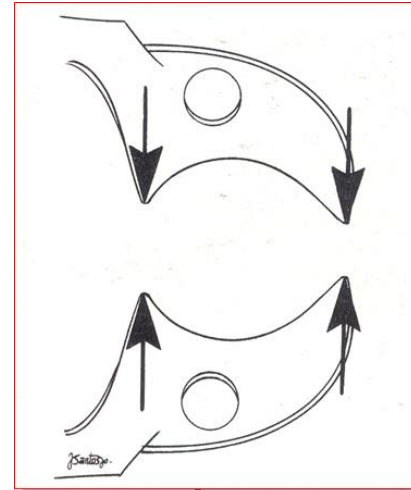
Rubber Dam Holder/frame

- It supports the edges of the rubber dam and thus improves access to the isolated teeth
- The Young frame is a U-shaped metal frame with small metal projections for securing the borders of the rubber dam
- Frame can be made of metal or plastic



Rubber dam retainer or clamp

- Rubber dam clamp consists of four prongs and two jaws connected by a bow.
- Primary function of rubber dam clamp is to anchor(stabilize) the rubber dam in position
- It is used to stabilize the dam to the most posterior tooth to be isolated.
- Many different sizes and shapes are available, with specific retainers designed for certain teeth



Clamps for different teeth

No. 7 - Mandibular molar

No. 8 - Maxillary molar

No. 4 - Most premolar,
especially upper

No. 2 - Small premolar like
first premolar anchor teeth

- Butterfly clamp – Anterior
teeth upper and lower

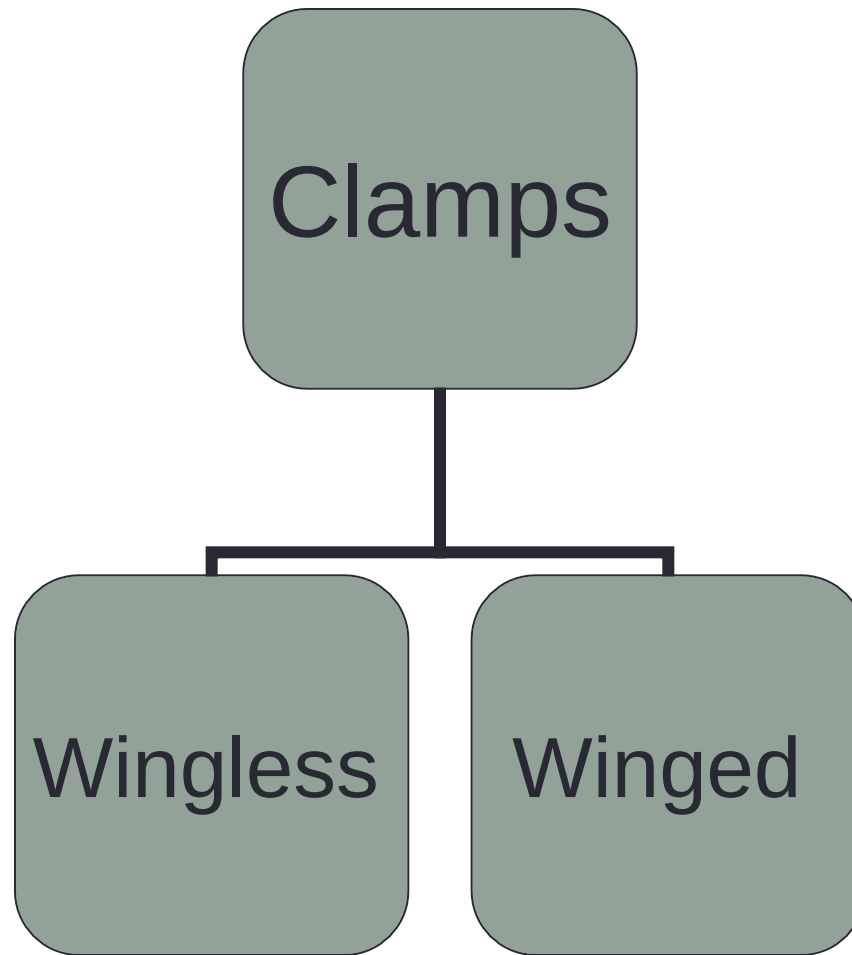
No. 212 – Cervical retainer
for Class V restoration



- When positioned on a tooth the properly selected retainer should contact the tooth in four areas, two on the facial surfaces and two on the lingual surface.
- This four point contact prevents rocking or tilting of the retainer.

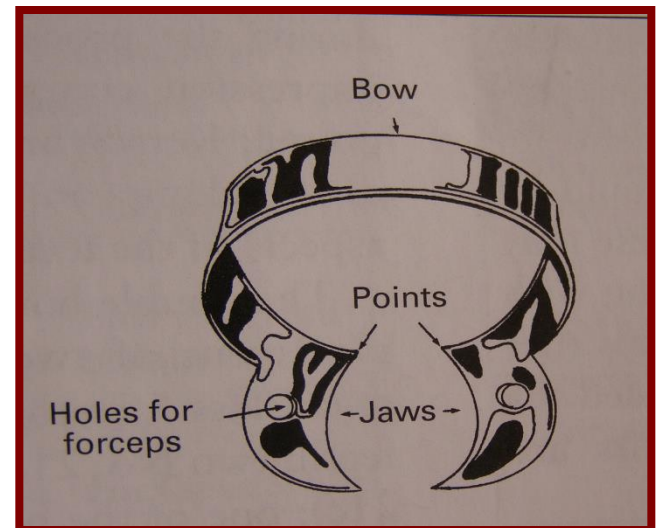
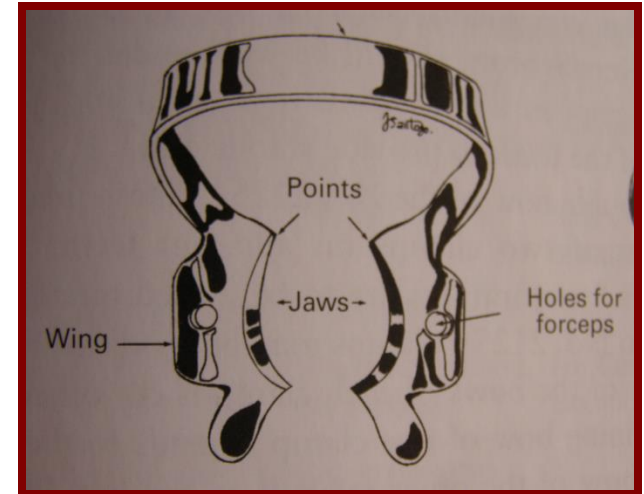
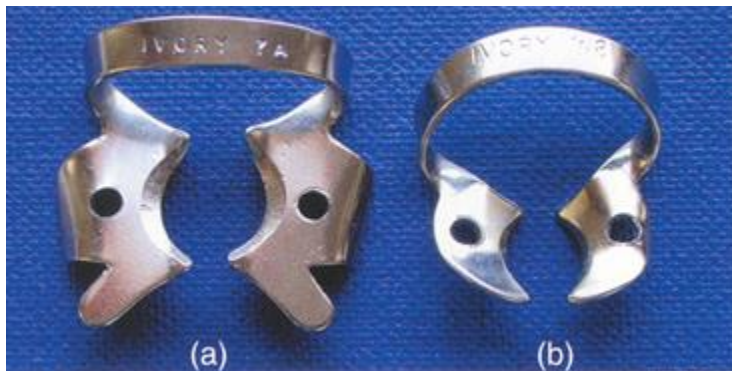


Retainer / Clamps



Wingless and Winged Clamp

- The winged retainer has anterior and lateral wings which provide extra retraction of the rubber dam from the operating field.
- A disadvantage of the winged retainers however interfere with the placement of matrix bands, band retainers and wedges and thus wingless retainers are preferred during Class II restorations



Anterior clamps

- **Butterfly clamps**

Small group of clamps have **two bows**, one on each end of the jaw, and due to their shape is called butterfly clamps.

A





FIESTA
COLOR
CODED CLAMPS





- The bow of the retainer should be tied with dental floss approximately 12 inches in length before the retainer is placed in the mouth.
- The floss allows retrieval of the retainer if the patient accidentally swallowed or aspirated.

CLAMP SWALLOWING/ ASPIRATION

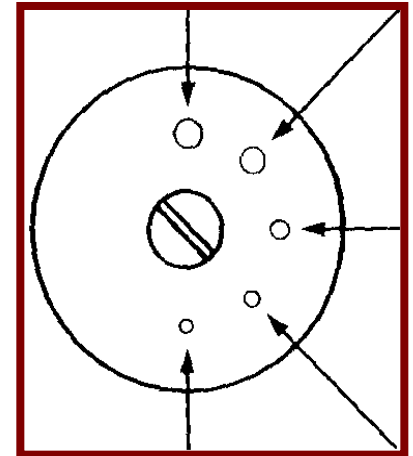
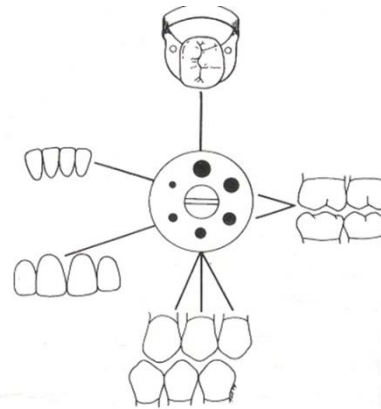
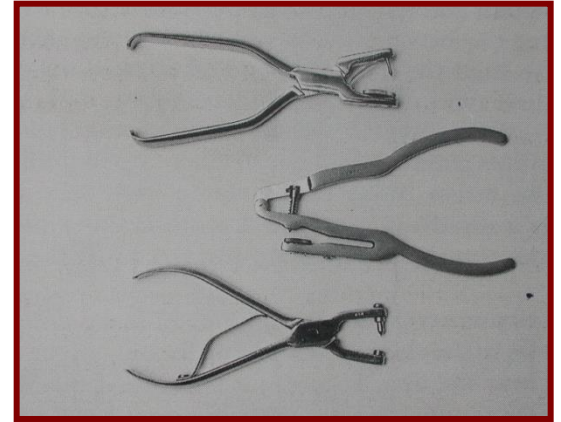


Various methods of tying retainers with dental floss

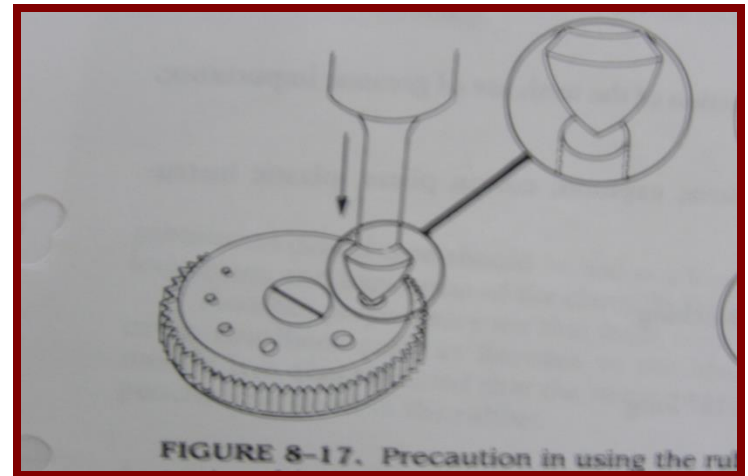


Rubber Dam Punch

- Holes in a range of sizes from 0.5-2.5mm in diameter
- Designs –
 - Ivory pattern
 - Ash or Ainsworth pattern

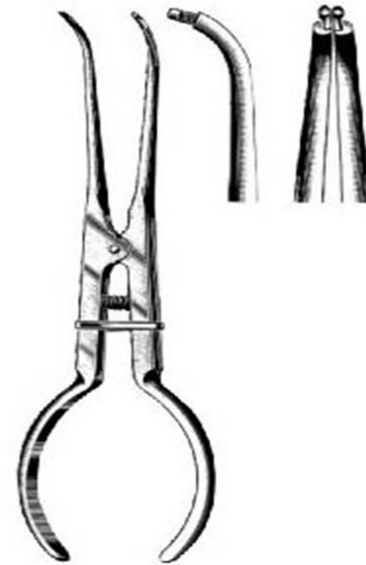


- It has a rotating metal disk with six holes of varying sizes and a tapered, sharp pointed plunger.

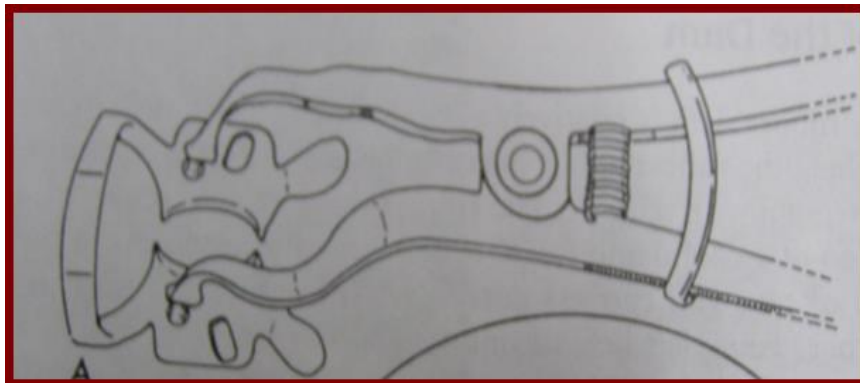


Rubber dam retainer forceps

- It is used both for the placement of the retainer and its removal from the tooth.
- Two widely used designs are
 1. Ash or Stokes
 2. Ivory pattern



Stokes



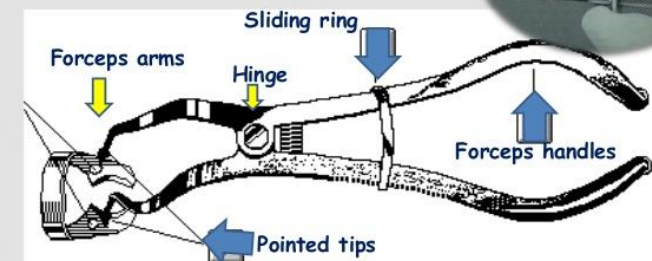
RUBBER DAM FORCEPS

Forceps are needed to stretch the jaws of the clamp open in a controlled manner during placement and removal.

Parts....

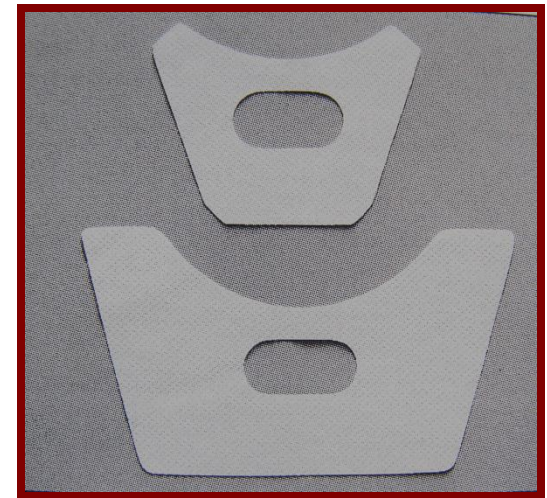
Sterilization.... Steam/ dry heat

Working life....

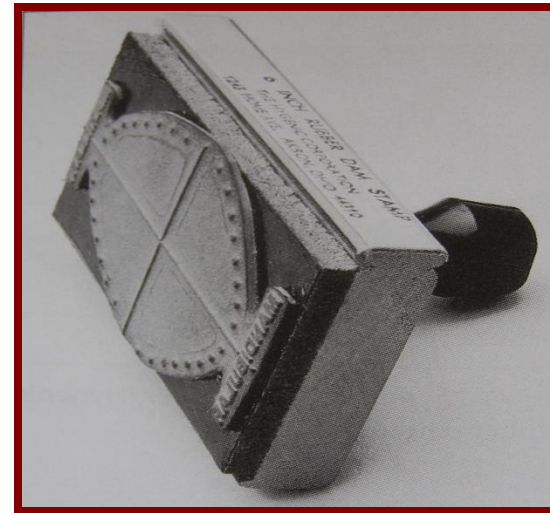


Rubber Dam Napkin

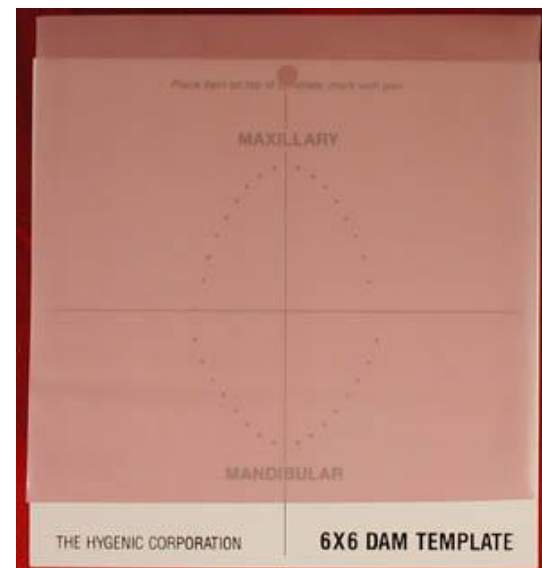
- Is placed between the rubber dam and patients skin.
- It prevents skin contact with rubber to reduce the possibility of allergic reactions.
- Absorbs saliva at the corners of the mouth.

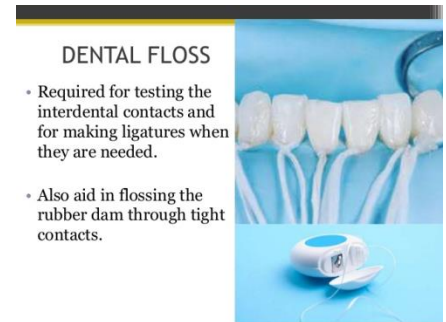


- A **rubber stamp** is available that imprints both permanent and primary arch forms in the rubber dam.



- A **plastic template** can also be used to mark hole positions.





- Anchors (Other than Retainers/clamps).

Sometimes Anchors other than retainers may be used.

To secure the dam further anteriorly or to anchor the dam on any tooth where a retainer is contraindicated, waxed dental tape or floss or a small piece of rubber dam material (cut from a sheet of dam) or a rubber Wedjet may be passed through the proximal contact.

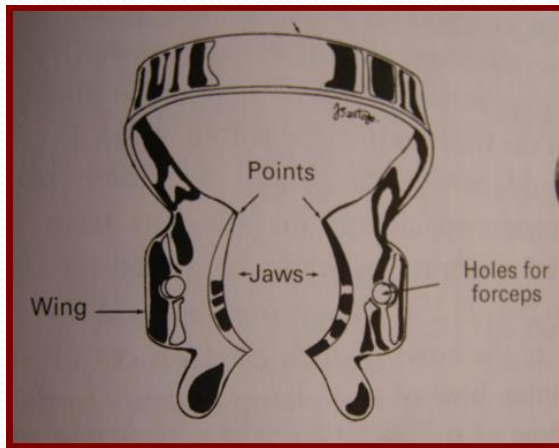


- **Technique I:** Clamp placement prior to Rubber dam

- As the clamp is placed first in this technique the tooth and gingival margin are clearly visible during placement and thus there is minimal risk of gingival trauma.



- **Technique 2 :** Clamp and rubber dam placed together.



- **TECHNIQUE 3:**

- **Clamp placement after the Rubber dam**

- This technique can prove difficult for an operator working alone and assistance will simplify the whole process.

- The disadvantage is the reduction in visibility of underlying gingival tissue, which may become impinged on by the retainer.



References

- Art and science of operative dentistry
- ✓ Sturdevant 5 th edition Page 463 – 468
- ✓ Sturdevant's 6 th edition art and science of operative dentistry, 189- 196